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Session Tracking Techniques

CSA4305– Internet Programming

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INTRODUCTION



- **HTTP is a stateless protocol**, meaning each request is treated independently by the web server.
- **Session tracking** is used to maintain a user's information across multiple requests.
- It helps web applications remember details such as **login status, user preferences, and shopping cart items**.
- Java web applications provide several session tracking mechanisms.
- The three important techniques are:
 - **HttpSession**
 - **Cookies**
 - **URL Rewriting**



What is Session Tracking?



- **HTTP Session** – Stores a user's session information on the **server**. It is secure and useful for applications like online shopping.
- **Cookies** – Stores small pieces of user information in the **browser**. They are easy to use but can be deleted or blocked by the user.
- **URL Rewriting** – Adds the session ID directly to the **URL** when cookies are unavailable. It works without cookies but can make URLs longer and expose session information.
- **Session Persistence** – This means keeping track of a user across multiple requests/pages. **HttpSession** provides good persistence, cookies can persist based on their expiry, and URL rewriting works as long as the session ID is passed in the URL.

Security & Use Cases – HttpSession is generally the **best choice for secure applications**. Cookies are convenient for remembering preferences/login information, while URL rewriting is mainly useful as a **backup when cookies are disabled**.



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Why Session Tracking Important!



- ❑ **Identifies Users** – Helps the server recognize the same user across multiple requests.
- ❑ **Maintains Login** – Keeps users logged in while they move between different pages.
- ❑ **Stores User Data** – Remembers information such as shopping cart items, preferences, and user choices.
- ❑ **Improves User Experience** – Allows websites to provide personalized and continuous experiences.
- ❑ **Supports Secure Transactions** – Helps maintain secure sessions for applications like banking, shopping, and online exams.
- ❑ **Example:**

Login → Browse Products → Add to Cart → Checkout



Http:



1. What is HTTP?

HTTP (HyperText Transfer Protocol) is a communication protocol used to transfer data between a **web browser and a web server**.

2. How it works

The browser sends an **HTTP request**, and the server sends back an **HTTP response**.

3. Why is it used?

It allows users to **access websites, web pages, images, videos, and other resources** over the internet.

4. Importance

HTTP is the basic protocol that enables **communication between clients and web servers**.



Advantage Http



- ☐ Session data is stored on the **server**.
- ☐ Can store multiple types of user information.
- ☐ Easy to use with Java Servlets.
- ☐ Suitable for login and authentication.
- ☐ Client does not directly store the actual session data.



Limitations



- ❑ Consumes server memory.
- ❑ Large numbers of users can increase server resource usage.
- ❑ Requires session management and expiration.
- ❑ Session ID theft can lead to session hijacking.



Cookies



What are Cookies?

Cookies are small pieces of data stored in the **user's browser** by a website.

Why are they used?

They help websites **remember user information**, such as login status, preferences, and shopping cart items.

How do they work?

The browser sends the cookie back to the server with future requests, helping the server **identify the user/session**.

Example

When you log in to a website and move between pages without logging in again, **cookies can help maintain your session**.



Engineer to Excel **Rewriting?**

URL REWRITING



It is a technique of adding **session information (like a session ID)** to the URL.

Why is it used?

It helps **track a user's session** when cookies are disabled or unavailable.

How does it work?

The session ID is added to the URL, for example: `website.com/page;sessionid=12345`.

Limitation

URLs become **longer**, and exposing the session ID in the URL can create **security risks**.

Advantages:

- Works even when **cookies are disabled**.



Session Persistence Comparison



- **HttpSession:** Session data is stored on the server and maintained using a session ID.
- **Cookies:** Store information in the browser and can persist until they expire or are deleted.
- **URL Rewriting:** Maintains the session by adding the session ID to the URL.



Security Concerns



- **HttpSession:** Generally more secure because session data stays on the server.
- **Cookies:** Can be vulnerable to theft or misuse, especially if not properly secured.
- **URL Rewriting:** Less secure because the session ID is visible in the URL and can be shared or exposed.



Use-Case Comparison



Method	Best Used For
HttpSession	Login sessions, shopping carts, secure web applications
Cookies	User preferences, remembering users, simple tracking
URL Rewriting	Session tracking when cookies are disabled



Conclusion



- Session tracking is essential because HTTP is stateless.
- **HttpSession** is suitable for maintaining server-side user sessions.
- **Cookies** are useful for storing small client-side information and session identifiers.
- **URL rewriting** provides an alternative when cookies are unavailable.
- Each mechanism has different advantages and security limitations.

THANK YOU